

Section - I

- 1.(b) Configuration of will be Cu^{2+} : $1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^9$
- 2.(d) The emission spectrum of hydrogen atom in visible region consists of a series of lines that are closer at higher energies.
- 3.(b) $PV = nRT$
Constant = n, T, R
 $P_1 V_1 = P_2 V_2$
 $10 \times 1 = 1 \times V_2$
 $V_2 = 10\text{ml}$
- 4.(a) For the given equilibrium $\text{H}_2\text{O}(s) \rightleftharpoons \text{H}_2\text{O}(l)$
By increasing pressure over the system melting point of ice will decrease and amount of water will increase.
- 5.(c) Acidic buffer having $\text{pH} = \text{pKa}$ has maximum buffer capacity.
- 6.(a) Cs is used in photoelectric cells because of low ionization enthalpy as electrons could be emitted very easily when exposed to light.
- 7.(a) Actually, PH_3 is formed there along with acetylene under the reaction of Ca_3P_2 with CaC_2 where, the phosphene reacts with acetylene formed and catches fire to get smoke which forms the smoke screen.
- 8.(d) Leaching method is used to concentrate the ores of gold, silver, aluminium, etc.
- 9.(b) C_2H_4 decolorizes alkaline KMnO_4 solution.
Alkaline KMnO_4 (Bayer's reagent) oxidizes ethylene to ethylene glycol and itself is reduced to MnO_2 .
- 10.(d)
- 11.(d)
- 12.(d) The electric lines of force cannot enter the metallic sphere as electric field inside the solid metallic sphere is zero. The electric field is always perpendicular to the surface of a conductor. On the surface of a metallic solid sphere, the electrical field is oriented normally (i.e. directed towards the centre of the sphere).
- 13.(b) $\text{KE} = \frac{GMm}{2r}$ i.e. $\text{KE} \propto r^{-1}$
& $\text{KE} \propto T^{-n}$ & $T^2 \propto r^3$
 $r \propto T^{2/3}$
Hence $r^{-1} \propto T^{-n}$
or, $r \propto T^n$
 $\therefore n = \frac{2}{3}$
- 14.(a) $A_1 v_1 = A_2 v_2$
or, $\frac{\pi d_1^2}{4} v_1 = \frac{\pi d_2^2}{4} v_2$
or, $2^2 \times v_1 = 4^2 \times v_2$
or, $v_1 = 4v_2$
- 15.(c) $v \propto r^2$
 $\therefore \frac{v'}{v} = \left(\frac{2r}{r}\right)^2 = 4$
 $\therefore v' = 4v$
Hence for $r' = 2r$
- 16.(a) Tension must be the same in both the rods for their junction to be in equilibrium.
 $Y_1 A \alpha_1 \theta = Y_2 A \alpha_2 \theta \Rightarrow Y_1 \alpha_1 = Y_2 \alpha_2$
- 17.(a) 2.54 contain the least significant figures of 3
 $97.52/2.54 = 38.393 = 38.4$
- 18.(a) $\frac{P_1}{T_1} = \frac{P_2}{T_2}$
or, $\frac{P_2}{P_1} = \frac{T_2}{T_1} = 2$
% increase = $\left(\frac{P_2}{P_1} - 1\right) \times 100\%$
 $= (2 - 1) \times 100\%$
 $= 100\%$
- 19.(b) $\frac{C}{100} = \frac{R - \text{LFP}}{\text{UFP} - \text{LFP}}$
or, $\frac{C}{100} = \frac{60 + 5}{95 + 5} = \frac{65}{100}$
 $\therefore C = 65^\circ\text{C}$
- 20.(b) $C = \frac{\epsilon_0 A}{d}$
 $C' = \frac{\epsilon_0 A}{d - \frac{d}{2}\left(1 - \frac{1}{\epsilon_r}\right)} = \frac{\epsilon_0 A}{d - \frac{d}{2}} = \frac{2\epsilon_0 A}{d} = 2C$
- 21.(d) Side of square loop, $l = 10\text{cm} = 0.1\text{m}$
Rate of change of magnetic field, $\frac{dB}{dt} = 1\text{ T s}^{-1}$
 $E = \frac{d\phi}{dt} = \frac{d(BAN)}{dt} = NA \frac{dB}{dt} = 500 \times 0.1^2 \times 1 = 5\text{V}$
- 22.(a) The charge passing through any cross section per unit time remain same i.e. current remain same.
- 23.(d) If the refractive index of the body becomes equal to surrounding liquid, there will not be any deviation in the direction of light neither will any light get reflected from its surface. So, the object becomes invisible.
- 24.(a) Interference in light waves is just a re-distribution of energy, depending on exactly which parts of the light wave overlap at each point in space. The pattern of light you see is then determined by the geometry.
- 25.(a) $P = I^2 R = \frac{E^2}{(R + r)^2} R$
E is constant and $(R + r)$ increases rapidly then $P \downarrow$
- 26.(d) $eV_s = hf - \phi$
- 27.(d) The forbidden energy gap in Ge is 0.7V.
- 28.(c) Let wire 1 lie along x-axis and wire 2 lie along y-axis. If at point (x,y) magnetic field is zero.
Then $\frac{\mu_0 i_1}{2\pi y} - \frac{\mu_0 i_2}{2\pi x} = 0 \Rightarrow y = \frac{i_1}{i_2} x$
Which is a straight line passing through the origin.
- 29.(d) $A = \{-2, 2\}$
 $B = \{x : x^2 = 4\}$
 $B = \{2, -2\}$
Hence, they are equal sets.
- 30.(a) $\cos\theta = \frac{(\vec{i} + \vec{j} + \vec{k}) \cdot (\vec{i} - \vec{j} - \vec{k})}{\sqrt{1^2 + 1^2 + 1^2} \cdot \sqrt{1^2 + (-1)^2 + (-1)^2}}$
 $= -\frac{1}{\sqrt{3}\sqrt{3}} = -\frac{1}{3}$
- 31.(d) If $x > 0$, then $\lim_{x \rightarrow 0^+} \frac{x}{x} = 1$
If $x < 0$, then $\lim_{x \rightarrow 0^-} \left(-\frac{x}{x}\right) = -1$
Since, RHL $\neq$ LHL limit does not exist.
- 32.(a) $y = \sinh^{-1}x + \cosh^{-1}x$
 $\frac{dy}{dx} = \frac{1}{\sqrt{1+x^2}} + \frac{1}{\sqrt{x^2-1}}$
- 33.(b) $\frac{dy}{dx} = 0$ $2x - 2 = 0$ $\therefore x = 1$
Putting $x = 1$, we get $y = \pm 2$

- 34.(d) Minimum value of $x = -\frac{b}{2a} = -\frac{8}{2} = -4$
 Minimum value = $16 - 32 + 17 = 1$
- 35.(a) $\cot^{-1}\left(\tan \frac{\pi}{7}\right) = \cot^{-1}\left(\cot\left(\frac{\pi}{2} - \frac{\pi}{7}\right)\right) = \frac{7\pi - 2\pi}{14} = \frac{5\pi}{14}$
- 36.(c) Skew symmetric matrix
- 37.(c) Inverse of a function exists if it is bijective.
- 38.(d) Since, $1^2 + (-1)^2 + 1^2 \neq 1$
- 39.(b)
- 40.(d) $\frac{dy}{dx} = \frac{d}{dx} \sin \frac{1}{x} = \frac{d \sin\left(\frac{1}{x}\right)}{d\left(\frac{1}{x}\right)} \frac{dx^{-1}}{dx} = -\frac{1}{x^2} \cos \frac{1}{x}$
- 41.(c) $\int \left(x + \frac{1}{x}\right) dx = \int x dx + \int \frac{1}{x} dx = \frac{x^2}{2} + \log x + c$
- 42.(a) $g = a, f = 0, c = 0$
 Radius = $\sqrt{g^2 + f^2 - c} = a$
- 43.(b) $x = \frac{t}{4}$
 or, $t = 4x \dots (i)$
 And, $y = \frac{t^2}{4}$
 $y = \frac{(4x)^2}{4} \quad x^2 = \frac{1}{4} y$
- It is a parabola
- 44.(d) Greatest side is 7
 The greatest angle is
 $\cos \theta = \frac{3^2 + 5^2 - 7^2}{2 \cdot 3 \cdot 5} = -\frac{1}{2} = \cos 120^\circ$
 $\therefore \theta = 120^\circ$
 So, it is obtuse angled Δ .
- 45.(b) $f(x) = \cos 4x + \tan 3x$
 $f(x + \pi) = \cos(4(x + \pi)) + \tan(3(x + \pi))$
 $= \cos(4x + 4\pi) + \tan(3x + 3\pi)$
 $= \cos 4x + \tan 3x$
 $\therefore$ The period of $f(x)$ is π .
- 46.(d) $A = \frac{G^2}{H}$
- 47.(c) $(1 - i)$
 Here, $r = \sqrt{1^2 + (-1)^2} = \sqrt{2}$
 $\tan \theta = \frac{x}{y} = -\frac{1}{1} = \tan(-45^\circ)$
 $\theta = -45^\circ$
 $\therefore (1 - i) = \sqrt{2} \{\cos(-45^\circ) + i \sin(-45^\circ)\}$
- 48.(d) $t_{n/2+1} = {}^nC_{n/2} a^{n/2} x^{n/2}$
 $= {}^{10}C_5 \left(\frac{y}{x}\right)^5 \left(-\frac{x}{y}\right)^5 = -252$
- 49.(c) 50.(b) 51.(d) 52.(d) 53.(a) 54.(c)
 55.(a) 56.(c) 57.(d) 58.(a) 59.(c) 60.(d)

Section - II

- 61.(b) $\text{Cu}^{2+} + 2\text{KI} \rightarrow \text{CuI}_2 + 2\text{K}^+$
 $2\text{CuI}_2 \rightarrow \text{Cu}_2\text{I}_2 + \text{I}_2$
- 62.(c) And in presence of acidic KMnO_4 alkene is oxidised to form acid.
 $\text{CH}_3 - \text{CH}_2 - \text{CH} = \text{CH} - \text{CH}_2 - \text{CH}_3 + (\text{H}_2\text{O} + \text{O}) \rightarrow 2\text{CH}_3\text{CH}_2\text{COOH}$

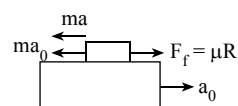
So the structure of X is
 $\text{CH}_3 - \text{CH}_2 - \text{CH} = \text{CH} - \text{CH}_2 - \text{CH}_3$
 Since it gives two moles of same carboxylic acid on reaction with oxidizing agent potassium permanganate hence it must be symmetrical alkene. Here the only alkene in option B is symmetrical alkene.

- 63.(c) $\text{Na}_2\text{S}_2\text{O}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{S} + \text{SO}_2 + \text{H}_2\text{O}$
- 64.(a) $\text{Ag}_2\text{CO}_3(\text{s}) \rightarrow 2\text{Ag}(\text{s}) + \text{CO}_2(\text{g}) + \frac{1}{2} \text{O}_2(\text{g})$
 Molecular weight of $\text{Ag}_2\text{CO}_3 = 276\text{g}$
 And, molecular weight of $\text{Ag} = 2 \times 108 = 216\text{g}$
 276g of Ag_2CO_3 give 216g Ag
 Therefore, 2.76g of Ag_2CO_3 on heating will give
 $\frac{216}{276} \times 2.76\text{g} = 2.16\text{g Ag}$ as residue.
- 65.(d) For neutral solution, $[\text{OH}^-] = \sqrt{K_w} = 10^{-8}$
 $\text{pOH} = -\log(10^{-8}) = 8$
- 66.(a) Meq of metal = Meq of HCl
 $\frac{0.9}{E} \times 1000 = 100 \times 1 \Rightarrow E = 9$
- 67.(c) The value of the Azimuthal quantum number always lies between 0 to $n - 1$. If $n = 4$, then $l = 4$ not possible.
- 68.(b) Up
 $h = -ut_1 + \frac{1}{2} gt_1^2 \dots (1)$

Down

- $h = ut_2 + \frac{1}{2} gt_2^2 \dots (2)$
- From (1) & (2)
 $-ut_1 + \frac{1}{2} gt_1^2 = ut_2 + \frac{1}{2} gt_2^2$
 or, $\frac{1}{2} g(t_1^2 - t_2^2) = u(t_2 + t_1)$
 or, $\frac{g}{2} (t_1 + t_2)(t_1 - t_2) = u(t_2 + t_1)$
 or, $x = \frac{2u}{g} = \frac{2 \times 10}{10} = 2 \text{ sec}$
- 69.(d) $F = \text{PA}$
 $= \frac{2T}{d} \times A = \frac{2 \times 70 \times 10^{-3} \times 10^{-2}}{0.05 \times 10^{-3}} = 28\text{N}$

70.(a)



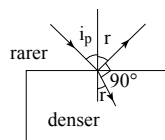
For block

$$ma_0 - F_f = ma_b$$

or, $m \times 4 - \mu mg = ma_b$
 or, $a_b = 4 - 0.2 \times 10 = 2 \text{ m/s}^2$
 $S_b = \frac{1}{2} a_b t^2 = \frac{1}{2} \times 2 \times 1^2 = 1 \text{ m}$ relative to ground.

- 71.(d) $T = \frac{1}{f} = 2\pi \sqrt{\frac{l}{g}} \Rightarrow f = \frac{1}{2\pi} \sqrt{\frac{g}{l}}$
 $\frac{f_1}{f_2} = \sqrt{\frac{l_2}{l_1}} \Rightarrow \left(\frac{n}{n+1}\right)^2 = \frac{l_2}{l_1} \quad \frac{l_1}{l_2} = \left(\frac{n+1}{n}\right)^2$

- 72.(d) $Q = nc_p dT$ for A
 $Q = mc_v dT$ for B
 $\therefore Q = nc_p \times 30 = nc_v \times dT$
 or, $dT = \gamma \times 30 = \frac{7}{5} \times 30 = 42\text{K}$
- 73.(b) Minimum frequency will be heard, when whistle moves away from the listener.
 $f_{\min} = f \left(\frac{v}{v + v_s} \right)$
 Where $v = r\omega = 0.5 \times 20 = 10 \text{ m/s}$
 $\Rightarrow f_{\min} = 385 \left(\frac{340}{340 + 10} \right) = 374 \text{ Hz}$
- 74.(c) $\bullet \rightarrow \bullet \rightarrow \bullet$
 Man Bus Catch
 $S_{\text{man}} - S_{\text{bus}} = 16$
 or, $vt - \frac{1}{2}at^2 = 16$
 or, $8t - \frac{1}{2} \times 2t^2 = 16$
 or, $t^2 - 8t + 16 = 0$
 or, $(t - 4)^2 = 0$
 $t = 4 \text{ sec}$
- 75.(c) 40% of $\frac{1}{2}mv^2 = msd\theta$
 or, $d\theta = \frac{0.4 \times 500^2}{2 \times 0.03 \times 4200} = 396^\circ\text{C}$
- 76.(d) 1st case
 $\frac{I}{I_0} = e^{-\mu x}$
 or, $\frac{1}{8} = \frac{1}{e^{\mu x}}$
 or, $e^{\mu x} = 8$
 or, $\mu x = \ln 8$
 or, $\mu = \frac{\ln 8}{27} \dots (i)$
2nd case
 $\frac{I}{I_0} = e^{-\mu x'}$
 or, $\frac{1}{2} = \frac{1}{e^{\mu x'}}$
 or, $e^{\mu x'} = 2$
 or, $\mu x' = \ln 2$
 or, $x' = \frac{\ln 2}{\ln 8} \times 27 = 9 \text{ mm}$
- 77.(c) $\mu = \tan i_p$
 $= \tan r$
 $\text{sinc} = \frac{1}{\mu} = \frac{1}{\tan r} = \cot r$
 $C = \sin^{-1}(\cot r)$
- 78.(b) $d \sin \theta_n = n\lambda$
 For max order $\sin \theta_n = 1$
 $d = n\lambda$
 $h = \frac{d}{\lambda} = \frac{700 \times 10^{-9}}{200 \times 10^{-9}} = 3.5 = 3$
 Total no. of maxima including central maxima
 $= 2n + 1 = 2 \times 3 + 1 = 7$



- 79.(c) First case
 $\frac{hc}{\lambda} = \phi + 6 \text{ eV}_0 \dots (1)$
2nd case
 $\frac{hc}{2\lambda} = \phi + 2 \text{ eV}_0 \dots (2)$
 From (1) & (2)
 $\frac{1}{2}(\phi + 6 \text{ eV}_0) = \phi + 2 \text{ eV}_0$
 or, $\phi + 6 \text{ eV}_0 = 2\phi + 4 \text{ eV}_0$
 or, $\phi = 2 \text{ eV}_0$
 From (1)
 $\frac{hc}{\lambda} = \phi + 3\phi$
 or, $\frac{hc}{\lambda} = 4 \frac{hc}{\lambda_0} \therefore \lambda_0 = 4\lambda$
- 80.(b) $i_c = \frac{90}{100} \times i_E \Rightarrow 10 = 0.9 \times i_E$
 $i_E = 11 \text{ mA}$
 Also, $i_E = i_c + i_B = 11 - 10 = 1 \text{ mA}$
- 81.(c) $E = E_5 - E_1$
 $= -\frac{13.6}{25} + 13.6$
 $= 13.056 \text{ eV}$
 Since momentum of photon = momentum of hydrogen atom.
 or, $\frac{E}{C} = mv$
 or, $v = \frac{13.056 \times 1.6 \times 10^{-19}}{3 \times 10^{-27} \times 1.67 \times 10^{-27}} = 4.2 \text{ m/s}$
- 82.(d) $f(x) = \frac{|x-2|}{x-2}$
 When $x > 2$, then $|x-2| = x-2$
 $f(x) = \frac{x-2}{x-2} = 1$
 When $x < 2$, then $|x-2| = -(x-2)$
 $4(x) = -\frac{(x-2)}{(x-2)} = -1$
 Range = $\{-1, 1\}$
- 83.(b) $|\vec{a}| = |\vec{b}| = |\vec{c}| = 1$
 We have, $\vec{a} + \vec{b} + \vec{c} = 0$
 $\vec{b} + \vec{c} = -\vec{a}$
 Squaring, $b^2 + 2\vec{b} \cdot \vec{c} + c^2 = a^2$
 $1^2 + 2bc \cos \theta + 1^2 = 1^2$
 $2\cos \theta = -1$
 $\therefore \cos \theta = -\frac{1}{2} \quad \text{i.e. } \theta = \frac{2\pi}{3}$
- 84.(d) $\int \frac{dx}{\cos x \sqrt{\cos 2x}} = \int \frac{\sec x}{\sqrt{\cos^2 x - \sin^2 x}} dx$
 $= \int \frac{\sec x \cdot dx}{\cos x \sqrt{1 - \tan^2 x}} = \int \frac{\sec^2 x}{\sqrt{1 - \tan^2 x}} \cdot dx$
 Let, $y = \tan x$
 $dy = \sec^2 x \cdot dx$
 $I = \int \frac{dy}{\sqrt{1 - y^2}} = \sin^{-1}(y) + c = \sin^{-1}(\tan x) + c$

85.(d) $\int |x| dx$
 Let $I = \int |x| \cdot 1 \cdot dx$

$$= |x| \int 1 \cdot dx - \int \left[\frac{d|x|}{dx} \int 1 \cdot dx \right] dx$$

$$= |x| \cdot x - \int \frac{x}{|x|} \cdot x \cdot dx$$

$$= |x| \cdot x - \int \frac{|x|^2}{(x)} dx \quad [\because x^2 = |x|^2]$$

$$I = |x| x - \int |x| dx$$

$$I = |x| x - I \quad \therefore I = \frac{|x|x}{2}$$

86.(b) Now, area bounded (A) = $\int_0^K x^2 dx$

$$= \int_0^{2^{1/3}} x^2 dx$$

$$\Rightarrow \frac{K^3}{3} = \frac{(2^{1/3})^3}{3} - \frac{K^3}{3}$$

$$\Rightarrow \frac{2K^3}{3} = \frac{2}{3} \quad \therefore K = 1$$

87.(d) Put $\theta - 15 = A$ and $\theta + 15 = B$
 i.e. $A + B = 2\theta$ and $A - B = 30$
 We have,

$$\frac{\tan A}{\tan B} = \frac{3}{1}$$

$$\frac{\tan A + \tan B}{\tan A - \tan B} = \frac{3+1}{3-1} \quad (\because \text{By component \& dividend})$$

$$\frac{\sin(A+B)}{\sin(A-B)} = 2$$

$$\Rightarrow \frac{\sin 2\theta}{\sin 30} = 2$$

$$\sin 2\theta = 2 \cdot \frac{1}{2} = 1$$

$$2\theta = (4n+1)\frac{\pi}{2} \quad \therefore \theta = n\pi + \frac{\pi}{4}$$

88.(a) $\cos \frac{A}{2} = \sqrt{\frac{b+c}{2c}}$
 or, $\cos^2 \frac{A}{2} = \frac{b+c}{2c}$

$$\frac{s(s-a)}{bc} = \frac{b+c}{2c} \cdot \frac{\left(\frac{a+b+c}{2}\right) \left(\frac{a+b+c}{2} - a\right)}{bc} = \frac{b+c}{2c}$$

 or, $\frac{(a+b+c)(b+c-a)}{2bc} = \frac{b+c}{c}$
 or, $(b+c)^2 - a^2 = 2b(b+c)$
 or, $b^2 + 2bc + c^2 - a^2 = 2b^2 + 2bc$

$$\therefore a^2 + b^2 = c^2$$

89.(b) $y = x + x^2 + x^3 + \dots + \infty$
 or, $y = \frac{x}{1-x}$
 or, $y - yx = x \Rightarrow x = \frac{y}{1+y}$

90.(b) Let α and $\alpha + 1$ be the roots. Then,

$$\alpha + \alpha + 1 = -a \Rightarrow \alpha = \frac{-a-1}{2}$$

and $\alpha(\alpha + 1) = c$
 or, $\alpha^2 + \alpha = c$
 or, $\left(\frac{-1-a}{2}\right)^2 + \frac{(-1-a)}{2} = c$
 or, $\frac{1+2a+a^2}{4} + \frac{(-1-a)}{2} = c$
 or, $\frac{1+2a+a^2-2-2a}{4} = c$
 or, $\frac{1+a^2-2}{4} = c$

$$\therefore a^2 = 4c + 1$$

91.(b) Total number of balls = $2 + 3 + 4 = 9$
 So, 3 balls can be drawn in 9C_3 ways.
 If black ball is neglected, total = 6 balls and 3 ball can be drawn in 6C_3 ways.
 So, total no. of ways that at least one is black ball

$$= {}^9C_3 - {}^6C_3$$

92.(b)
$$\frac{(1-i)^3}{1-i^3} = \frac{(1-i)^3}{1^3-i^3} = \frac{(1-i)^3}{[(1-i)(1+i+i^2)]}$$

$$= \frac{(1-i)^2}{(1+i-1)} = \frac{(1-i)^2}{i} = \frac{1+i^2-2i}{i}$$

$$= \frac{(1-1-2i)}{i} = -\frac{2i}{i} = -2$$

93.(c) $2p = \left| \frac{1}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2}}} \right| \Rightarrow \frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{4p^2}$

$$\Rightarrow \frac{1}{a^2} + \frac{1}{b^2} = \frac{2}{8p^2} \quad \therefore \frac{1}{a^2}, \frac{1}{8p^2}, \frac{1}{b^2} \text{ are in A.P.}$$

94.(a) Let $A(1, 1)$ in ΔABC
 $(-2, 3)$ is mid-point of $AB \Rightarrow B = (-5, 5)$
 $(5, 2)$ is mid-point of $AC \Rightarrow C = (9, 3)$
 Centroid of $\Delta ABC = \left(\frac{1-5+9}{3}, \frac{1+5+3}{3} \right) = \left(\frac{5}{3}, 3 \right)$

95.(b) According to given conditions only internal touch is possible.

Hence, $C_1 C_2 = |r_1 - r_2| \Rightarrow \frac{c}{2} = \left| \frac{c}{2} - 2 \right|$

Taking +ve, $\frac{c}{2} = \frac{c}{2} - 2$ (Not possible)

Taking -ve, $\frac{c}{2} = -\frac{c}{2} + 2$

$$c = 2$$

96.(a) Put $x = 0$
 $y^2 - 5y + 6 = 0 \quad \therefore y = 2 \text{ or } 3$
 Parabola intersect of y-axis at $(0, 2)$ or $(0, 3)$
 So, intercept = 1

97.(a) 98.(d) 99.(a) 100.(b)

...The End...